

Forensic Investigation Report

Case ID	CASE_2CF24DBA5F	Report ID	RPT-2CF24DBA5F-C186CB39
Generated (UTC)	2026-07-25T13:02:00.398Z	Report version	v2

Every statement in this report references stored forensic findings; no content is generated outside computed results. Section "Report Provenance & Integrity" lists the SHA-256 digests of the source artifacts.

Executive Summary

- Case CASE_2CF24DBA5F contains 4 evidence item(s), each acquired under SHA-256 chain-of-custody verification.
- This case is linked to 1 other case(s) through shared entities: CASE_8F43434664 [cross_case_correlation.json].
- The weighted correlation engine found 6 related evidence pair(s) out of 6 analysed [correlation_analysis.json].
- The strongest suspect anchor is 'security@nabil-bank-alerts.com' (emails) with confidence 39/100 [suspect_assessment.json].
- Observed attack progression: initial_contact -> financial_transaction -> social_engineering -> credential_theft [timeline_analysis.json].

Scope & Methodology

- **objective:** Acquire, verify, correlate and reconstruct the digital evidence for this case, and derive investigative leads (suspect anchors, campaigns, cross-case links) strictly from stored, hash-verified artifacts.
- **evidence scope:** 4 evidence item(s) acquired through the CIIS intake pipeline under SHA-256 chain-of-custody control.
- **methodology:**
 - Phase 1 - Acquisition & OCR: PaddleOCR PP-OCRv5 text extraction with per-item confidence scoring; SHA-256 fingerprint recorded at intake and re-verified at read.
 - Phase 1 - Forensics: metadata/EXIF consistency, forgery signals, logo detection and evidence-confidence scoring stored per item under storage/forensics/.
 - Phase 2 - Correlation: weighted entity-overlap engine scoring every evidence pair; results in correlation_analysis.json.
 - Phase 2 - Cross-case: shared-entity matching against every other analysed case (cross_case_correlation.json).
 - Phase 2 - Campaigns / Suspects / Timeline: clustering, anchor derivation and event reconstruction over the correlated evidence set.
 - Threat intelligence: URL/domain indicators scored by the configured provider (static indicator file, or the trained phishing classifier when CIIS_ML_THREAT_INTEL=1); per-indicator results in 'Model Prediction Results'.
 - Reporting: this document is assembled exclusively from the stored outputs above; it contains no free-text generation.
- **reproducibility:** Re-running the analysis against the same stored evidence reproduces every figure herein; artifacts are versioned and never overwritten.

Case Overview

- **case id:** CASE_2CF24DBA5F
- **evidence count:** 4
- **first evidence:** 2026-07-25T12:50:00.695Z
- **last evidence:** 2026-07-25T12:53:47.011Z
- **file types:**
 - jpg
 - png
 - txt

Evidence Summary

Evidence	File	Acquired	OCR conf.	Integrity
EVID_00001	yeti.jpg	2026-07-25T12:50:00	92%	VERIFIED
EVID_00002	whatsapp_chat_export.txt	2026-07-25T12:53:07	100%	VERIFIED
EVID_00003	romanchat.jpg	2026-07-25T12:53:23	97%	VERIFIED
EVID_00004	phishing_email_screenshot.png	2026-07-25T12:53:47	99%	VERIFIED

SHA-256 digests for each item are listed in the Appendix (chain of custody).

Correlation Analysis

- **pair count:** 6
- **related pair count:** 6
- **strength distribution:**
 - **WEAK:** 6
- **top relationships:**
 - **pair:** EVID_00001 <-> EVID_00002
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (yeti.jpg) and EVID_00002 (whatsapp_chat_export.txt) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.1 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00001 <-> EVID_00003
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (yeti.jpg) and EVID_00003 (romanchat.jpg) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.1 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00001 <-> EVID_00004
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (yeti.jpg) and EVID_00004 (phishing_email_screenshot.png) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.1 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00002 <-> EVID_00003
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00002 (whatsapp_chat_export.txt) and EVID_00003 (romanchat.jpg) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00002 <-> EVID_00004
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00002 (whatsapp_chat_export.txt) and EVID_00004 (phishing_email_screenshot.png) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].

Cross-Case Correlation

- **related case count:** 1
- **related case ids:**
 - CASE_8F43434664
- **links:**
 - **other case id:** CASE_8F43434664
 - **relationship strength:** VERY_STRONG

- **match confidence:** 0.8262
- **match reason:** Shares 4 entity(ies) with CASE_8F43434664: domain nabil-bank-alerts.com, domain nabil-verify.example-scam.top, email security@nabil-bank-alerts.com (+1 more)
- **matched entities:**
 - **entity type:** domains
 - **value:** nabil-bank-alerts.com
 - **this evidence ids:**
 - EVID_00004
 - **other evidence ids:**
 - EVID_00006
 - **entity type:** domains
 - **value:** nabil-verify.example-scam.top
 - **this evidence ids:**
 - EVID_00004
 - **other evidence ids:**
 - EVID_00006
 - **entity type:** emails
 - **value:** security@nabil-bank-alerts.com
 - **this evidence ids:**
 - EVID_00004
 - **other evidence ids:**
 - EVID_00006
 - **entity type:** urls
 - **value:** http://nabil-verify.example-scam.top/login
 - **this evidence ids:**
 - EVID_00004
 - **other evidence ids:**
 - EVID_00006

Campaign Analysis

- **campaign count:** 0
- **unclustered evidence:**
 - EVID_00001
 - EVID_00002
 - EVID_00003
 - EVID_00004
- **campaigns:**
 - none

Timeline Analysis

- **summary:** 4 event(s) spanning 4851.7 hour(s); 0 timestamp(s) unresolved. Observed scam progression: initial_contact -> financial_transaction -> social_engineering -> credential_theft.
- **stage progression:**
 - initial_contact
 - financial_transaction
 - social_engineering
 - credential_theft
- **progression consistent:** False
- **milestones:**
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **description:** Investigation start - first reconstructed evidence event
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **description:** First observation of stage 'initial_contact' (EVID_00002)

- **timestamp:** 2026-01-04T09:12:00+00:00
- **description:** First observation of stage 'financial_transaction' (EVID_00002)
- **timestamp:** 2026-07-25T12:53:47.011000+00:00
- **description:** First observation of stage 'social_engineering' (EVID_00004)
- **timestamp:** 2026-07-25T12:53:47.011000+00:00
- **description:** First observation of stage 'credential_theft' (EVID_00004)
- **critical events:**
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **evidence id:** EVID_00002
 - **reasons:**
 - contains money entity/entities: Rs 2000

Suspect Assessment

- **suspect id:** SUSPECT_CASE_2CF24DBA5F_01
- **identity:** emails:security@nabil-bank-alerts.com
- **confidence score:** 38.8
- **confidence level:** LOW
- **risk level:** LOW
- **evidence ids:**
 - EVID_00004
- **explanation:** Suspect anchor 'security@nabil-bank-alerts.com' (emails) scores 38.8/100 (LOW, risk LOW) across 1 evidence item(s): EVID_00004. Identity strength: 75/100 (weight 0.25) - 'security@nabil-bank-alerts.com' is a email - identity weight 75/100 for this anchor type. Evidence count: 25/100 (weight 0.20) - appears in 1 of 4 evidence item(s). Evidence confidence: 50/100 (weight 0.15) - mean Phase-1 evidence confidence of its evidence set is 50/100 (no Phase-1 scores stored; neutral 50 assumed). Threat intelligence: 0/100 (weight 0.15) - no threat-intelligence indicator file available. Correlation strength: 50/100 (weight 0.15) - fewer than two evidence items (or no correlation input); neutral 50. Timeline span: 0/100 (weight 0.10) - activity spans 0.0 day(s) across its evidence set.

Threat Intelligence Summary

No threat-intelligence indicator file was provided; threat corroboration was skipped (not an absence of threat).

Model Prediction Results

No threat-intelligence provider was active during this analysis; indicator-level model predictions were not produced (this is a coverage gap, not an absence of threat).

Evidence Quality Summary

- **mean image quality:** 0.0
- **mean evidence confidence:** 0.0
- **mean forgery score:** 0.0
- **max forgery score:** 0.0
- **mean ocr confidence:** 0.97
- **hash verified count:** 4.0

Metadata Summary

- **note:** No Phase-1 metadata reports stored for this case.

Investigation Statistics

- **entity statistics:**
 - **dates:** 1
 - **domains:** 2
 - **emails:** 1
 - **money:** 1

- ports: 2
- times: 5
- urls: 1
- campaign statistics:
 - campaign count: 0.0
 - largest campaign size: 0.0
 - clustered evidence: 0.0
 - unclustered evidence: 4.0
 - mean campaign confidence: 0.0
- timeline statistics:
 - event count: 4.0
 - resolved event count: 4.0
 - unresolved event count: 0.0
 - inferred event count: 3.0
 - stage count: 4.0
 - critical event count: 1.0
 - timeline span hours: 4851.7
- correlation statistics:
 - pair count: 6.0
 - related pair count: 6.0
 - mean confidence: 0.2212
 - max confidence: 0.2212

Confidence Analysis

- **note:** No Phase-1 confidence scores stored for this case.

Investigation Conclusion

- 4/4 evidence item(s) passed SHA-256 chain-of-custody verification.
- The evidence set is internally connected (6 weighted relationship(s)), consistent with related activity rather than isolated incidents.
- Investigation should focus on anchor 'security@nabil-bank-alerts.com' (39/100 confidence).
- Observed stage order deviates from the canonical scam sequence; evidence acquisition order should be reviewed.

Recommendations

- Prioritise the 1 critical event(s) involving OTP/financial entities for victim-impact assessment.
- Case priority: LOW (13.8/100) - Low urgency: archive-ready unless new evidence raises the correlation or threat picture.

Report Provenance & Integrity

- **report id:** RPT-2CF24DBA5F-C186CB39
- **generated at:** 2026-07-25T13:02:00.398Z
- **generator:** CIIS Phase-2 reporting module (template-over-data; no free-text generation)
- **evidence set digest:** 845d2580a93c7505d88a3b735dd3a956db233d26f779d05f4a85f60abfe82c3f
- **evidence set digest note:** SHA-256 over the sorted SHA-256 digests of every evidence item; any change to the evidence set changes this value.

Source artifact digests (SHA-256):

correlation_analysis_v5.json: 606cd3e9ddfb9ea51edae3fd4809b28c145ded3c31c71a77552d3134012b22d5
 cross_case_correlation_v2.json: 5c99bf3ca3349a165019bf94ec27b048b011f080ca66bcc3c8944c4b5b845494
 campaign_analysis.json: 69e9ef6d9e9ed939484307d8d50cb975803a505a9aeb0a7926fde19b38bb344e
 suspect_assessment.json: 2eb4374c1f785a610e6d940e9cf2d9e038cc6e33a958bca933ba2e0ddb89da88
 timeline_analysis_v5.json: d79e22ff86d5f7eb147dcc7d438e8e13adeedea98c801ae754f82f67ddb2dfd5
 analytics.json: 22df3eb99a31dff38b24892ee9adfdb7fe951595e96e99232962fc231313ee4d
 case_priority.json: 186dbbf2d52b4faa5c8d137af1e04e7d5c86110faa5c4a680f482ff16d93b2b5
 graph_v5.json: dc41844c51c4135f888e6339c7f06cfc362878b86373ddedbcff39f112f94327

Appendix

- **chain of custody:**
 - **evidence id:** EVID_00001
 - **sha256:** 8f2e60ea6fc0eeffd209e28eb29228797104b60a2505934fc3d42642c100ea79
 - **upload time:** 2026-07-25T12:50:00.695Z
 - **status:** processed
 - **evidence id:** EVID_00002
 - **sha256:** c7ab347cfbd2b80c944c85b3442f3b2c400da1d3b92962f9ded8598e1e06cfb2
 - **upload time:** 2026-07-25T12:53:07.182Z
 - **status:** processed
 - **evidence id:** EVID_00003
 - **sha256:** 0340d5863313f4cc0b9b40ed835a77a71b8304d2e08eb32f08a56ed93a6e0b10
 - **upload time:** 2026-07-25T12:53:23.174Z
 - **status:** processed
 - **evidence id:** EVID_00004
 - **sha256:** 8ef0f27a84960b656d9051892e36096ce56fe9f30eb9a935d4d4330a002b9b7f
 - **upload time:** 2026-07-25T12:53:47.011Z
 - **status:** processed
- **stored artifacts root:** storage/investigation/<CASE_ID>/
- **phase1 artifacts root:** storage/forensics/<EVIDENCE_ID>/

Prepared by (system)

Cybercrime Investigation Intelligence System
Automated Phase-2 reporting module

Reviewed by (investigator)

Date:

Date:

Signature: